



Medical Computer Vision for the Clinic: Gallbladder Cancer and Beyond

Presented By: Dr. Soumen Basu

06.10.2026

About Me

Education

- PhD (IIT Delhi, 2024)
- M.Tech (IIT Delhi, 2015)
- BE (IEST, Shibpur, 2012)

Employment

- *Assistant Professor*, Department of AI, IIT Kharagpur (2026 – Present)
- *Research Scientist*, Model Scalability Innovations, Meta New York (2025 – 2026)
- *Senior Chief Engineer*, Visual Intelligence Team, Samsung Research Bangalore (2024 – 2025)
- *Member of Tech Staff*, AdClouds, Adobe Systems Bangalore (2015 – 2018)

Areas of Interest

- Computer Vision, Deep Learning, AI in Healthcare



THE PLEDGE

Trivia: Indian Healthcare – Challenges

- Universal Health is not a reality despite years of planning and investment – Healthcare in India is still fragmented
- 80% of specialists concentrate on metropolitan areas – Rural India sees 70% shortage of specialists (UHM, 2024)
- Rural residents have only 13% access to primary health centers, 33% to subcenters, and 9.6% to hospitals [a]
- Tier-2 and Tier-3 cities lack adequate advanced machines – CT and MRI scanners are scarce or outdated, pushing patients toward metros for diagnostic care

[a] <https://pmc.ncbi.nlm.nih.gov/articles/PMC10446776/>

Medical Computer Vision in Practice

1 Noisy data

Ultrasound speckle, shadows and spurious textures. Handheld, operator-dependent scans where the view changes from one sweep to the next.

2 Domain shift

Different machines, hospitals and patient populations. A model that is accurate at one centre can quietly fail at the next.

3 Diagnostic trust

A clinician must know when to rely on a prediction and when to override it. Accuracy alone does not earn that.

This talk: what one end-to-end clinical pipeline (gallbladder cancer) taught us, and where it is taking the lab next.

Roadmap of This Talk

GBC detection from ultrasound

- GBCNet, US-UCL, FocusMAE
- Interpretability
- Prospective study and deployment

Limitations

- What breaks when we scale to multi-center deployment

CVI Lab: current research threads

- Adaptive, open-vocabulary, efficient and trustworthy models
- Open problems for new scholars





THE TURN

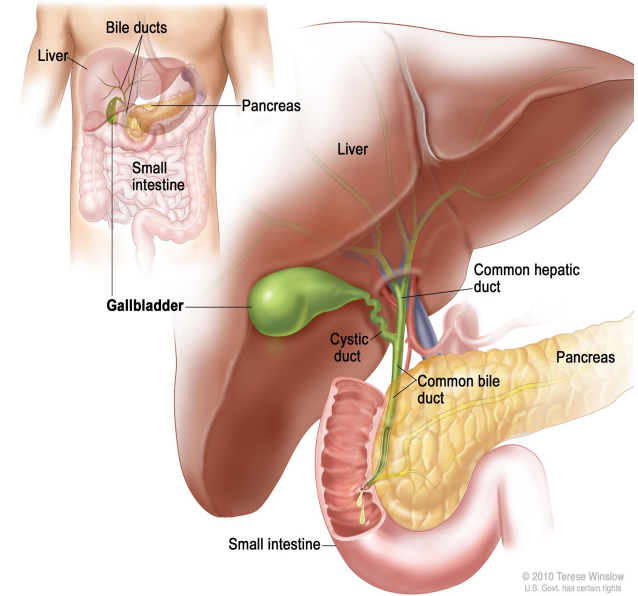


Gallbladder Cancer Detection from Ultrasound

Clinical reality, handheld ultrasound, and models that made it to the hospital

Gallbladder Cancer (GBC)

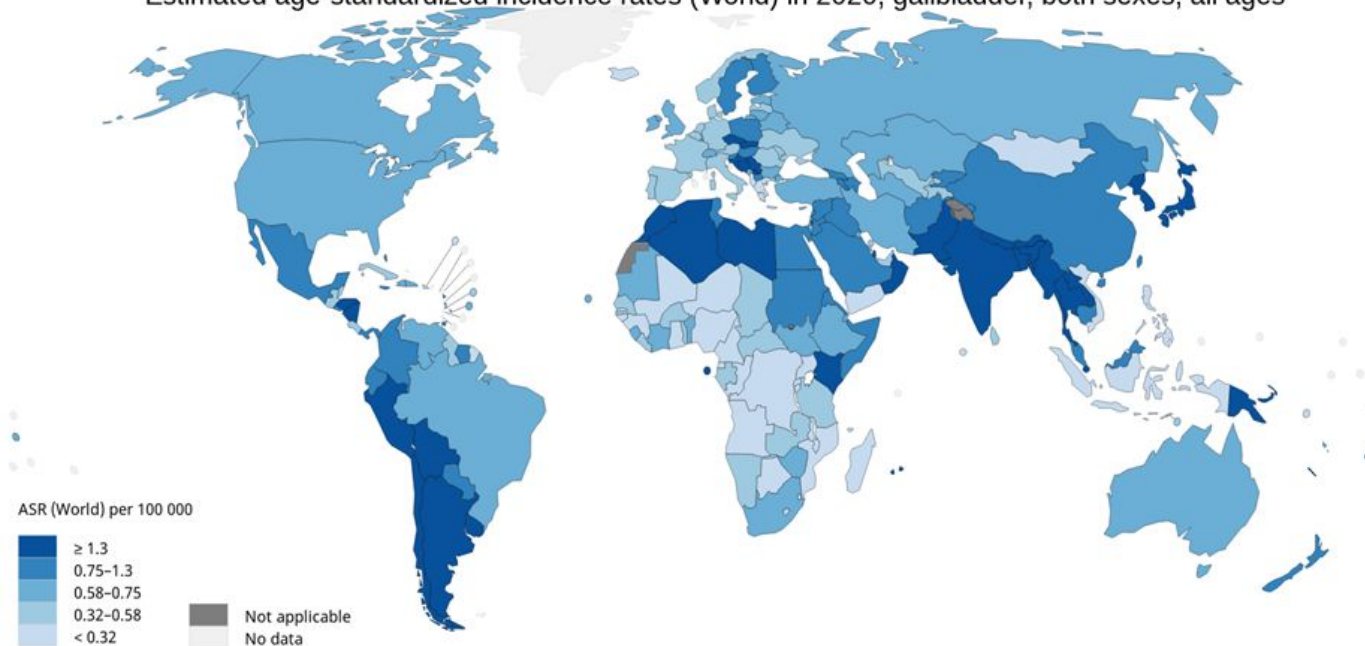
- Worldwide - 84,000 deaths every year [1]
- 5 year survival rate is 5%
Mean survival - 6 months
- Quick Metastasis - contiguous liver tissues
- Silent progress - often detected at a very late stage



[1] GLOBOCAN 2020 data

India - Higher Risk

Estimated age-standardized incidence rates (World) in 2020, gallbladder, both sexes, all ages



Why Ultrasound (US) Imaging for GBC?

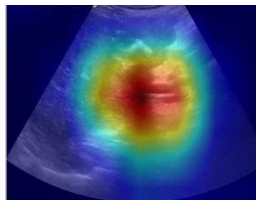
- Most common imaging modality for abdominal ailments – often the sole diagnostic imaging performed
- Highly accessible and low cost (compared to CT/ MRI which are high cost, and only available in handful places in low resource countries) – excellent candidate modality for GBC detection
- No work on AI/ML-based GBC detection from US prior to ours

Motivation

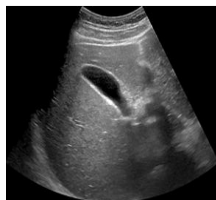
- Timely detection can increase 5 year survival rate to 44% [1]
- Non-expert radiologists perform poorly on US images
- Expert radiologists showed about ~70% sensitivity while differentiating between benign and malignant GBs in US images (blinded from other clinical data)
- AI-based automated detection for second reading – improve accuracy, triage

[1] Chen et al. *Long-term outcomes and prognostic factors in advanced gallbladder cancer: focus on the advanced t-stage*. PloS one, 2016

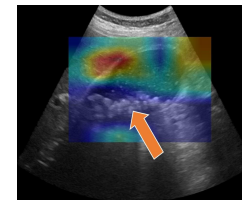
Challenges



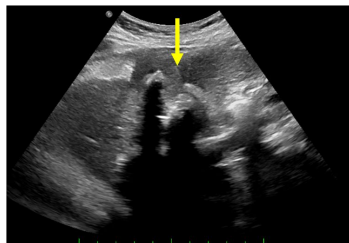
Low image quality - sensor artifacts (noise, shadow) – models underperform



Difficulty due to absence of clear anatomical structure



Bias to spurious textures from noise and adjacent organs – increase false positive



Due to the handheld sensor, the scanning plane may change – introduce observer bias

Left and right shows same GB from different scanning views – drastic change

Surpassing the Human Accuracy: Detecting Gallbladder Cancer from USG with Curriculum Learning

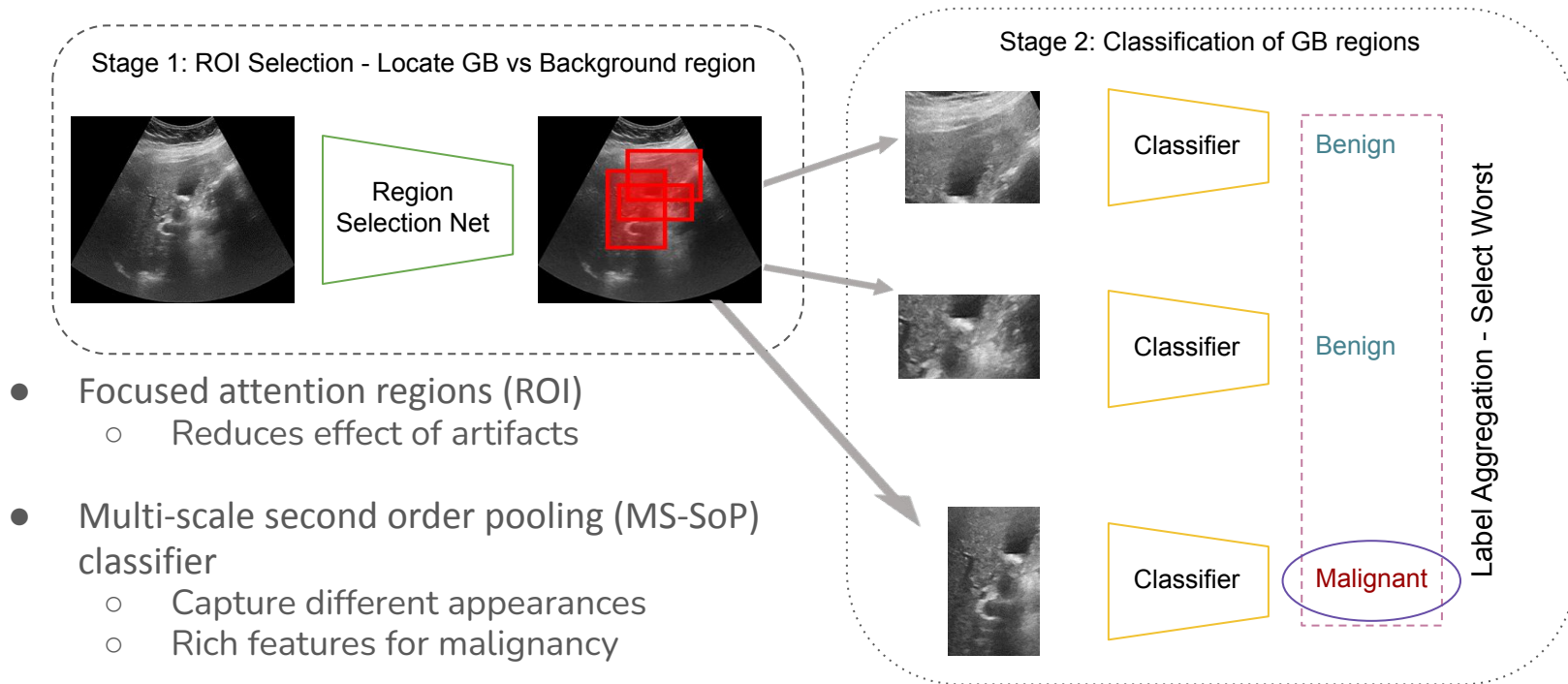
Soumen Basu, Mayank Gupta, Pratyaksha Rana, Pankaj Gupta, Chetan Arora
Computer Vision and Pattern Recognition (CVPR) 2022



Project Page: <https://gbc-iitd.github.io/gbcnet>

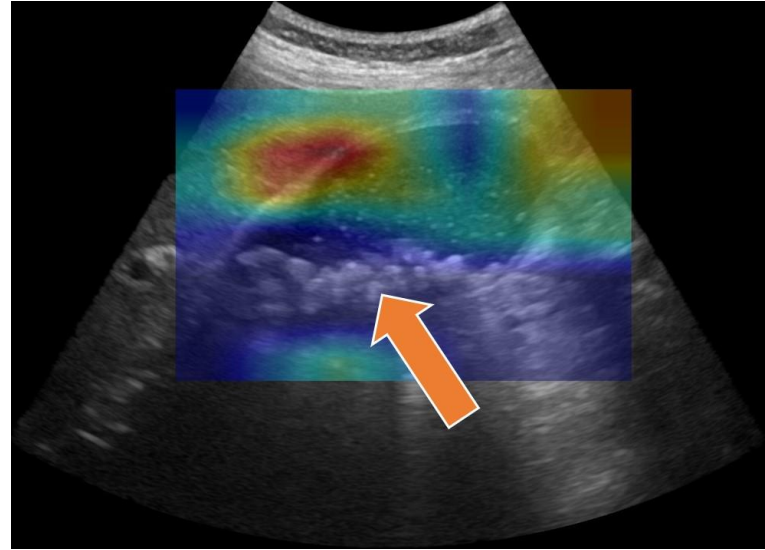
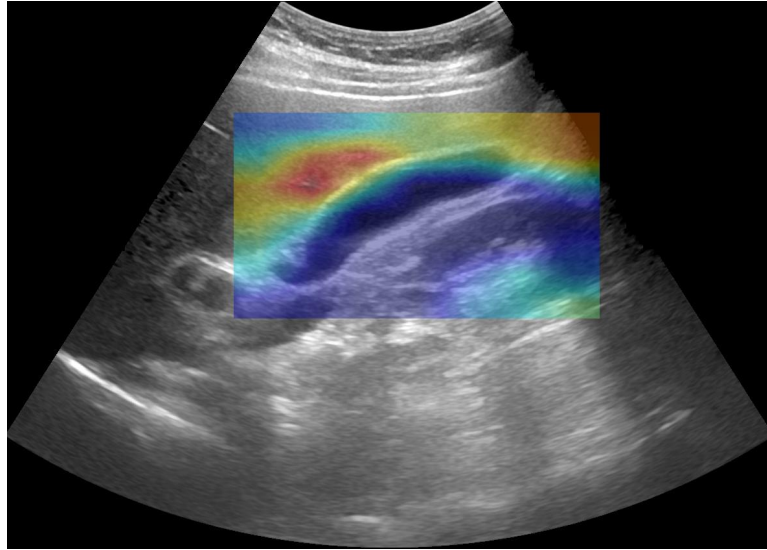


Our Solution: GBCNet

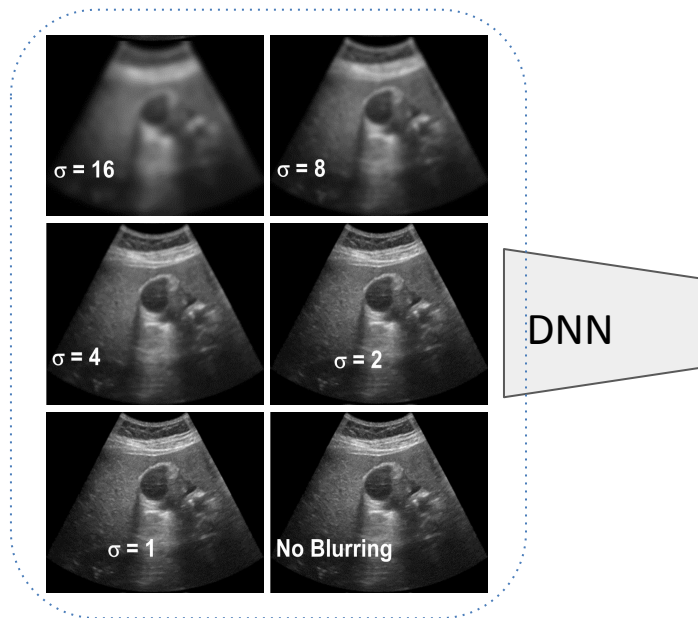


Issues with Textures

- Spurious echogenic textures of adjacent organs may bias the network decision
- Increased false positives (non-malignant GB being predicted as malignant)



Visual Acuity-based Curriculum - Tackle Texture Bias



Algorithm 1: Proposed Visual Acuity-based Curriculum

Input: D^{train} , Dataset of regions cropped from the original USG images.

Output: Optimized model parameters W^*

Initialize $\sigma = \sigma_0$;

Initialize model parameters to W ;

for $epoch = \{1 \dots, N\}$ **do**

if $\sigma > 0$ **then**

$Z = \phi$;

for $x \in D^{\text{train}}$ **do**

$Z = Z \cup \{x \circledast G(\sigma)\}$;

end

 train(W, Z);

else

 train(W, D^{train}) ;

end

if ($epoch > k'$) and ($epoch \% k == 0$) **then**

$\sigma = \lfloor \sigma / 2 \rfloor$;

end

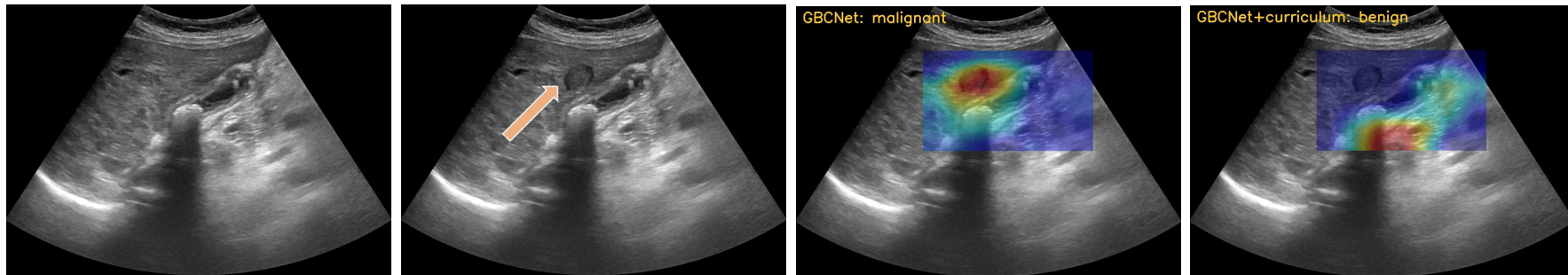
end

- Gaussian Blurring to simulate visual acuity and used as a training curriculum
- Start training with blurred images - gradually introduce high resolution images

Effect of Curriculum in Tackling Textures

Synthetic data - added textures and tissue-like patches in the non-malignant samples (without altering the GB pathology)

Curriculum trained networks are not susceptible to such textures - results in lower false-positives



Results – Effect of Curriculum

Model	Spec.	Sens.
ROI+VGG16	0.719 ± 0.115	0.733 ± 0.179
ROI+VGG16+VA	0.938 ± 0.030	72.0 ± 0.195
ROI+ResNet50	0.823 ± 0.105	0.909 ± 0.111
ROI+ResNet50+VA	0.923 ± 0.059	0.875 ± 0.091
ROI+Inception-V3	0.833 ± 0.087	0.785 ± 0.214
ROI+Inception-V3+VA	0.931 ± 0.044	0.826 ± 0.099
GBCNet (ROI+MS-SoP)	0.942 ± 0.037	0.923 ± 0.071
GBCNet+VA	0.967 ± 0.023	0.919 ± 0.063

Results – Comparison w/ Baselines

Method	Test Set			Cross Val.		
	Acc.	Spec.	Sens.	Acc.	Spec.	Sens.
Radiologist A	0.816	0.873	0.707	–	–	–
Radiologist B	0.784	0.811	0.732	–	–	–
VGG16	0.721	0.900	0.381	0.693 ± 0.036	0.960 ± 0.046	0.495 ± 0.234
ResNet50	0.787	0.875	0.619	0.867 ± 0.070	0.926 ± 0.069	0.672 ± 0.147
Inception V3	0.850	0.875	0.801	0.844 ± 0.039	0.953 ± 0.029	0.807 ± 0.097
Faster-RCNN	0.779	0.762	0.810	0.757 ± 0.053	0.840 ± 0.046	0.808 ± 0.104
RetinaNet	0.836	0.863	0.786	0.749 ± 0.073	0.867 ± 0.078	0.791 ± 0.089
EfficientDet	0.779	0.863	0.620	0.739 ± 0.084	0.881 ± 0.099	0.858 ± 0.061
GBCNet	0.910	0.900	0.929	0.882 ± 0.051	0.942 ± 0.037	0.923 ± 0.071
GBCNet+VA	0.959	0.950	0.976	0.921 ± 0.029	0.967 ± 0.023	0.919 ± 0.063

Unsupervised Contrastive Learning of Image Representations from Ultrasound Videos with Hard Negative Mining

Soumen Basu, Somanshu Singla, Mayank Gupta, Pratyaksha Rana, Pankaj Gupta, Chetan Arora
Medical Image Computing and Computer Assisted Intervention (MICCAI) 2022

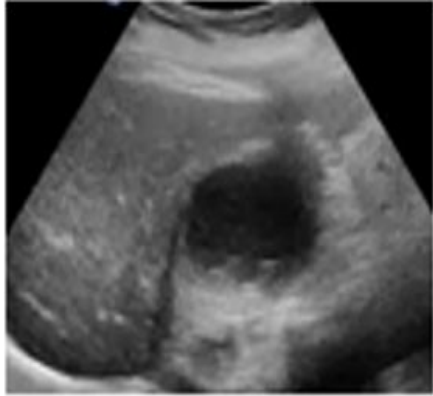


Project Page: <https://gbc-iitd.github.io/usucl>



Key Idea

Frame: T



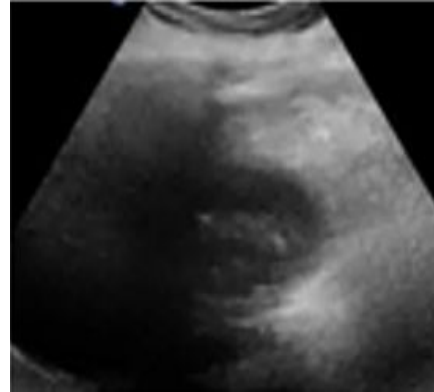
Malignant
Sample with
stones and wall
thickening

T+2



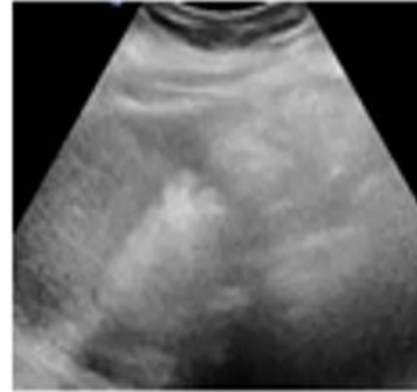
Positive Pair of the
sample also
showing wall
thickening and
stones

T+t1



“Hard” Negative
sample showing
the GB but wall
thickening is not
visible

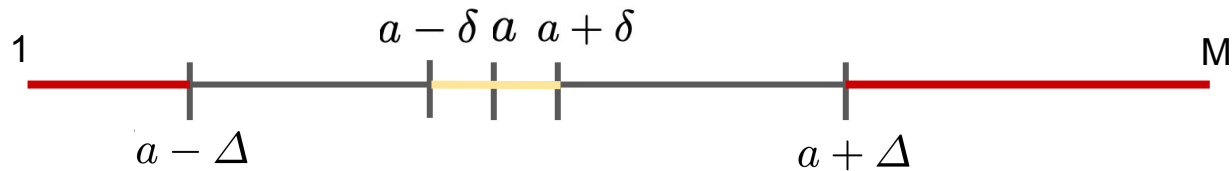
T+t2



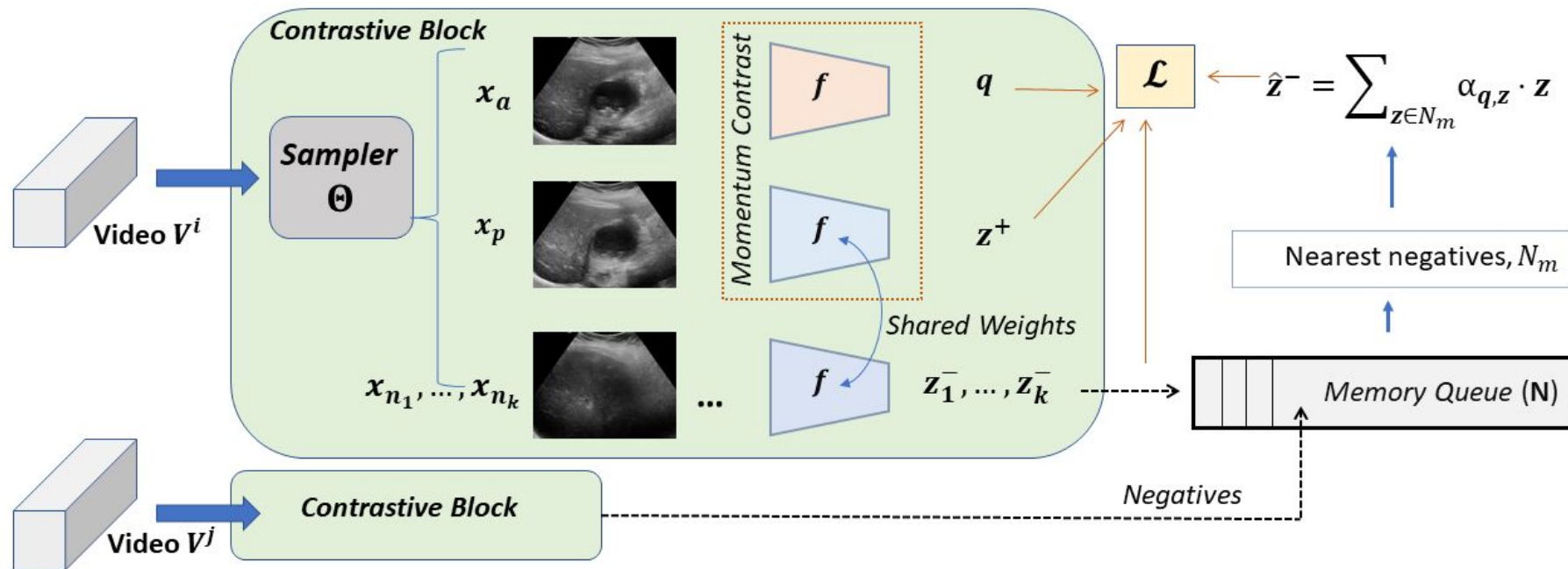
“Easy” Negative
where even the
GB is not visible

Video Sub-Sampling

- $\mathbf{V} = \{\mathbf{x}_j\}_{j=1}^M$
- $\Theta : \mathbf{V} \rightarrow (\mathbf{x}_a, \mathbf{x}_p, \{\mathbf{x}_{n_1}, \dots, \mathbf{x}_{n_k}\})$
- $a \sim U([1, M])$, $p \sim U([a - \delta, a + \delta] \setminus \{a\})$,
 $n_1, \dots, n_k \stackrel{\text{i.i.d.}}{\sim} U([1, M] \setminus [a - \Delta, a + \Delta])$



Method - Pretraining Overview



Contrastive Loss Function

- Normalized similarity measure between anchor and cross-video negatives

$$\alpha_{\mathbf{q}, \mathbf{z}} = \frac{\exp(s(\mathbf{q}, \mathbf{z})/\tau)}{\sum_{\mathbf{z}_c \in N} \exp(s(\mathbf{q}, \mathbf{z}_c)/\tau)}$$

Cosine Similarity: $s(\mathbf{a}, \mathbf{b}) = \mathbf{a} \cdot \mathbf{b} / (\|\mathbf{a}\|_2 \|\mathbf{b}\|_2)$

- Rank top-n cross-video negatives based on $\alpha_{\mathbf{q}, \mathbf{z}}$
- Cross-video negative (aggregated) $\hat{\mathbf{z}}^- = \sum_{\mathbf{z} \in N_m} \alpha_{\mathbf{q}, \mathbf{z}} \mathbf{z}$.

$$\mathcal{L} = -\log \frac{\exp(s(\mathbf{q}, \mathbf{z}^+)/\tau)}{\underbrace{\exp(s(\mathbf{q}, \mathbf{z}^+)/\tau) + \sum_{j=1}^k \exp(s(\mathbf{q}, \mathbf{z}_j^-)/\tau)}_{\text{intra-video}} + \underbrace{\exp(s(\mathbf{q}, \hat{\mathbf{z}}^-)/\tau)}_{\text{cross-video}}}$$

Key Results - GBC

- Results for ResNet50 classifier model with different pre-training

Method		Acc.	Spec.	Sens.
Pretrained on ImageNet		0.867 ± 0.070	0.926 ± 0.069	0.672 ± 0.147
Pretrained on GB Videos	SimCLR [10]	0.897 ± 0.040	0.912 ± 0.055	0.874 ± 0.067
	SimSiam [11]	0.900 ± 0.052	0.913 ± 0.059	0.861 ± 0.061
	BYOL [15]	0.844 ± 0.129	0.871 ± 0.144	0.739 ± 0.178
	MoCo v2[19]	0.886 ± 0.061	0.893 ± 0.078	0.871 ± 0.094
	Cycle-Contrast [25]	0.861 ± 0.087	0.867 ± 0.098	0.844 ± 0.097
	USCL [12]	0.901 ± 0.047	0.923 ± 0.041	0.831 ± 0.072
Ours		0.921 ± 0.034	0.926 ± 0.043	0.900 ± 0.046

FocusMAE: Gallbladder Cancer Detection from Ultrasound Videos with Focused Masked Autoencoders

Soumen Basu, Mayuna Gupta, Chetan Madan, Pankaj Gupta, Chetan Arora
Computer Vision and Pattern Recognition (CVPR) 2024

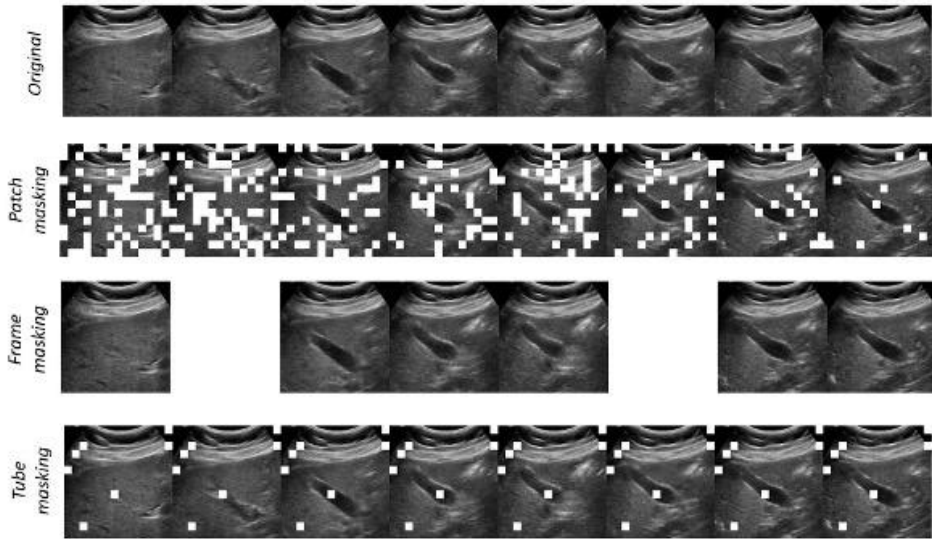


Project Page: <https://gbc-iitd.github.io/focusmae>



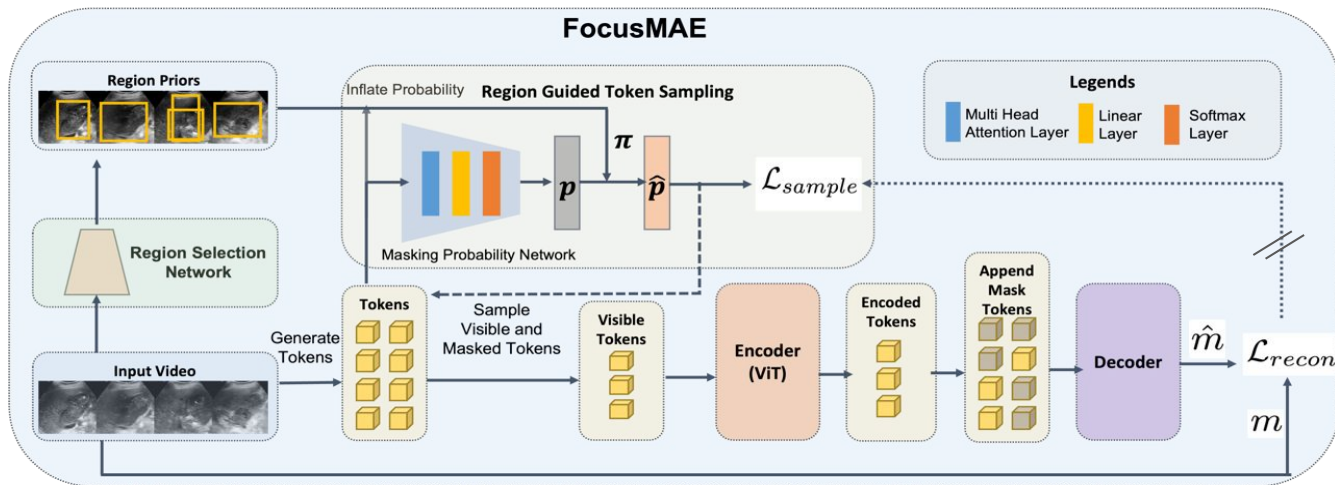
Why MAE with Random Masking is Suboptimal

- MAE variants predominantly use random masking of patches
- In our case, regions indicative of malignancy typically constitute a small portion – "information density" is highly non-uniform
- Random masking is not robust to small pathology areas with large background (low info) regions – learns the background representation
- Solution – Focused Masking



Various common random masking strategies

Our Solution: FocusMAE – Overview

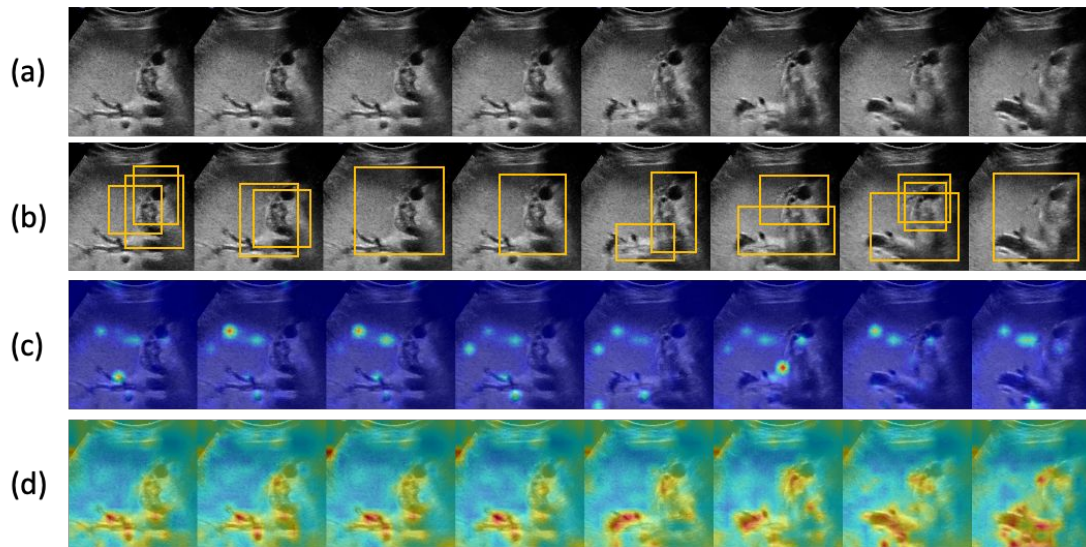


- Use Object Detector to generate high information region priors (candidate ROI)
- Bias the masking probability of the tokens within ROI to learn representation of the pathology/ disease

Key Results

Group	Method	Backbone	Acc.	Spec.	Sens.
Human Experts	Radiologist A	–	0.786±0.134	1.000±0.000	0.672±0.201
	Radiologist B	–	0.874±0.088	1.000±0.000	0.811±0.126
Image-based	ResNet50 [25]	CNN	0.711±0.091	0.822±0.102	0.672±0.147
	InceptionV3 [43]	CNN	0.734±0.089	0.953±0.072	0.647±0.107
	Faster-RCNN [41]	CNN	0.757±0.058	0.687±0.056	0.808±0.091
	EfficientDet [44]	CNN	0.789±0.084	0.761±0.099	0.828±0.061
	ViT [13]	Transformer	0.796±0.068	0.751±0.128	0.820±0.076
	DEiT [46]	Transformer	0.829±0.034	0.787±0.154	0.845±0.058
	PVTv2 [49]	Transformer	0.831±0.041	0.857±0.167	0.834±0.068
	GBCNet [5]	CNN	0.840±0.105	0.843±0.204	0.843±0.072
	US-UCL [8]	CNN	0.808±0.127	0.871±0.217	0.776±0.109
	RadFormer (SOTA) [6]	Transformer	0.840±0.105	0.776±0.162	0.877±0.088
Video-based	Video-Swin [34]	Transformer	0.925±0.053	1.000±0.000	0.903±0.085
	TimeSformer [9]	Transformer	0.920±0.058	0.967±0.067	0.909±0.058
	VidTr [33]	Transformer	0.924±0.038	1.000±0.000	0.800±0.072
	VideoMAEv2 [48]	Transformer	0.942±0.066	0.937±0.078	0.940±0.120
	AdaMAE [4]	Transformer	0.947±0.053	0.952±0.066	0.913±0.116
	FocusMAE (Ours)	Transformer	0.964±0.047	0.910±0.117	1.000±0.000

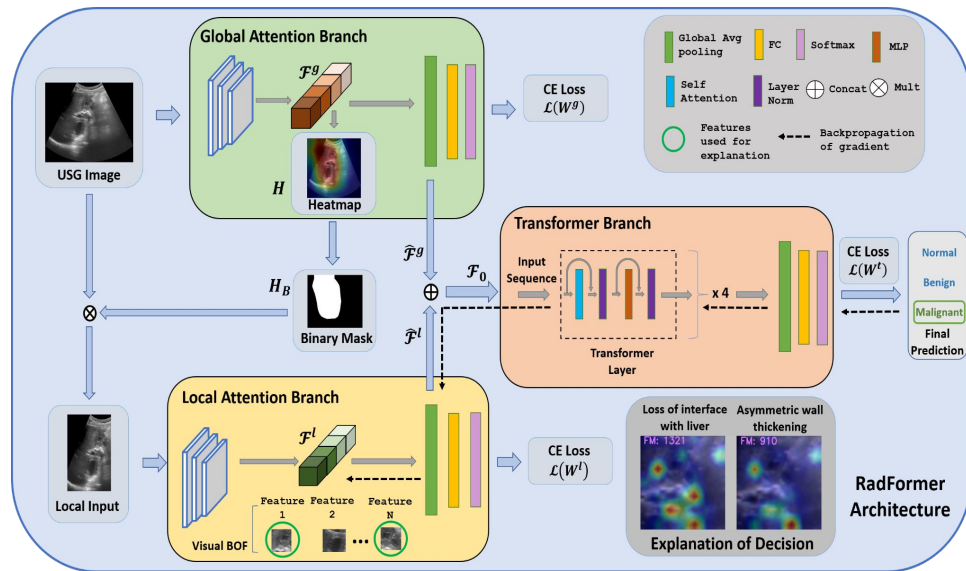
Qualitative Analysis



- FocusMAE (d) attentions are more guided to the pathology and anatomical structures as compared to VideoMAE (c) attentions.

Interpretability [*Med. Image Analysis, 2023, IF 14.0*]

- Global-local attention based model
- Local branch - visual bag-of-feature style embedding
- Top visual words (gradient-weighted) are mapped with radiological features
- Explainable unit features consistent with RADS - compose radiologist-like explanations



Neural Features vs. RADS Lexicons Mapping

Algorithm 1: Lexicon and neural feature mapping

Input: Set of images and associated lexicon labels,
 $X = \{I_i, L_i\}_{i=1}^N$, where $L_i \in \mathcal{P}(\{l_1, \dots, l_k\})$. $\mathcal{P}$ is the powerset and l_j are individual lexicons.

Output: Dictionaries R and M of key-value pairs, where in R , keys are lexicons and values are corresponding neural features. In the reverse map M , key: neural feature, value: corresponding lexicon.

1. Initialize set of lexicon labels $L = \phi$;
2. Set of all distinct lexicons $D = \phi$;
3. Set of lexicons that do not appear individually in the images $C = \phi$;
4. Initialize dictionaries $S = \{\}$, $R = \{\}$, $M = \{\}$;

for $i=\{1, \dots, N\}$ **do**

F_i, W_i = Get features with gradient weights for I_i ;

T_i = Find-Top-Feature-IDs(W_i) ;

$S[L_i] = S[L_i] \cup T_i$;

$L = L \cup \{L_i\}$;

$D = D \cup L_i$;

end

foreach $l \in D$ **do**

if $l \in S.Keys()$ **then**

$R[l] = S[l]$;

else

$C = C \cup \{l\}$;

end

end

foreach $x \in L$ **do**

foreach $y \in L$ **do**

if $(x \setminus y \in C)$ **and** $(C \neq \phi)$ **then**

$R[x \setminus y] = S[x] \setminus S[y]$;

$C = C \setminus \{x \setminus y\}$;

end

end

end

foreach $(K, V) \in R$ **do**

foreach $feature, f \in V$ **do**

$M[f] = K$;

end

end

return R, M ;

Function Find-Top-Feature-IDs(W):

$k = \text{argmax}\{W\}$ and $W^* = W_k$;

Initialize $T = \{k\}$;

for $j=\{1, \dots, |W| \setminus \{k\}\}$ **do**

if $|W^* - W_j| \leq \epsilon$ **then**

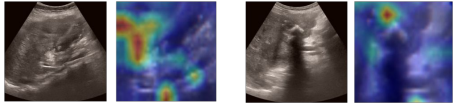
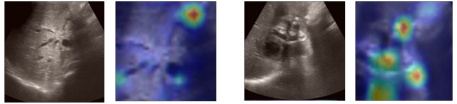
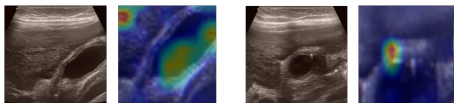
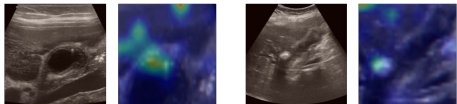
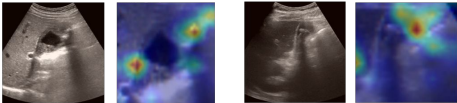
$T = T \cup j$;

end

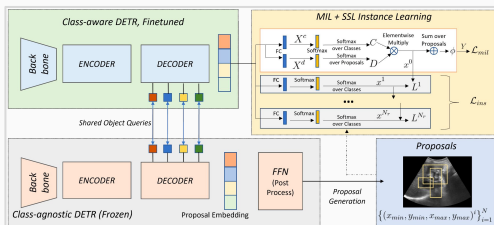
end

end

return T ;

RadFormer feature id	GB-RADS lexicons	Sample images with activation visual of the feature from local BOF
1321	Loss of interface between GB wall and liver –significantly associated with malignancy	
1955, 1807, 1581, 1657	Extramural invasion – significantly associated with malignancy	
1935, 1359	Mural layering – intact inner and outer layers of the GB wall, favoring benign pathology	
879	Intramural echogenic mural foci – due to cholesterol deposition/ intramural calcification	
876	Intramural cyst within the GB wall – evidence of Rokitansky-Aschoff sinuses (RAS)	

Other Works

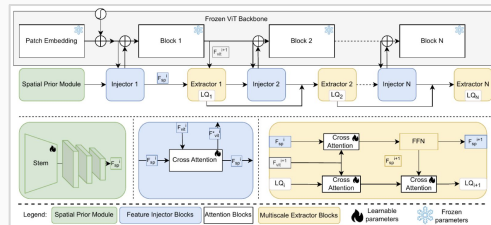


Weakly Supervised GBC Detection

MICCAI 2023

DETR adapted to learn from image-level labels only, removing the need for bounding-box annotation.

AP 0.628 vs 0.531 for prior SOTA (WS-DETR)

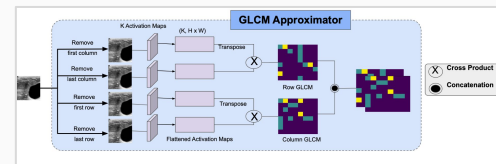


LQ-Adapter

WACV 2025, Oral

ViT-Adapter with learnable queries, a foundation-model route to GBC detection from ultrasound.

Outperforms DINO and DETR detectors on GBC



GLCM-MAE

MICCAI 2025

MAE pre-training with a differentiable GLCM-style loss that keeps subtle diagnostic textures.

+2.1% GBC (US), +3.1% breast (US), +0.5% CXR, +0.6% CT

Also: XGC vs. GBC differentiation on ultrasound (Indian J. Gastroenterol. 2024); AI in biliary tract cancers, review (Indian J. Gastroenterol. 2024)

Prospective Study [*Lancet Reg. Health 2024, IF: 6.2*]

- Large prospective study – MS-SoP DL model for automatic detection of GBC at abdominal US
- Compare diagnostic performance with two expert radiologists.
- Perform extensive subgroup analysis to demonstrate the robustness of DL-model.
 - Polyps and mural thickening,
 - Contracted gallbladder
 - Neck lesions, etc.
- 565 prospective patients at PGIMER – temporally different data cohort from GBCNet data
 - Train: 233, Val: 59, Held out Test: 273

GBC Detection – Deployment

Computer Vision Demo Projects
Project Demonstration of Computer Vision Group CSE IIT Delhi. Logout

Gall Bladder Cancer Detection System

Choose File

no file selected

Your Prediction:

Normal

Ground Truth:

Not Available

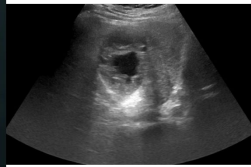
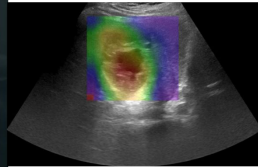
Select Model:

Resnet50

☐ // We give consent to IIT Delhi to store this image for non-profit research purposes

Submit

Computer Vision Demo Projects
Project Demonstration of Computer Vision Group CSE IIT Delhi. Logout

Uploaded Image	Prediction	CAM / Grad-CAM Visual
	Malignant	
	Predicted Probabilities	
	Normal : 0.34 Benign : 0.1 Malignant : 0.56	

Try Another Image

Download Results

- Cloud App, and Jetson-based Edge Deployment
- Clinical Testing initiated for real-time patients at PGIMER, AIIMS

Limitations

Domain shift

What happens when the ultrasound machine changes?

Annotation

How do we handle rare, overlapping pathologies we haven't explicitly annotated?

Trust

How does a clinician know when to trust the model's boundary prediction?

Specialised, single-centre models do not scale. The next generation has to be unified and adaptive.



THE PRESTIGE



CVI Lab Research Threads

Unified, adaptive models for the next generation of medical CV

Bottlenecks and Research Threads

Domain shift



Generative test-time adaptation

Adapt to a new hospital's scans on the fly, without retraining

Compute Efficiency



Training-Free Adaptive-step diffusion

Make high-resolution generative synthesis viable on edge devices

Annotation



Open-vocabulary tasks

Long-text clinical descriptions guide the model zero-shot

Trustworthy



Segmentation/ Detection with uncertainty guarantees

Probabilistic latent autoregression of aleatoric uncertainty

Challenge 1: Adapting to the Wild

OUR DIRECTION

Generative augmentation for test-time adaptation. The model adapts to each new hospital's data distribution on the fly, using only the unlabelled scans it sees, with no retraining and no patient data leaving the site.

OPPORTUNITIES

- One model across many machines and hospitals
- Deployment without a new annotation round per site

OPEN PROBLEMS YOU COULD OWN

- Adapting reliably from only a handful of test scans
- Knowing when adaptation helps and when it hurts
- Extending from images to ultrasound video

Q. How do we handle domain shifts across hospitals without retraining?

Challenge 2: Computational Efficiency

OUR DIRECTION

Training-free, adaptive-step diffusion sampling. The sampler spends compute only where an image needs it, so high-resolution generative synthesis, and the adaptation it powers, runs on edge hardware like our Jetson deployment.

OPPORTUNITIES

- Generative adaptation inside a rural clinic
- No retraining of the underlying generative model

OPEN PROBLEMS YOU COULD OWN

- Fewer sampling steps without losing diagnostic detail
- Fitting edge memory and latency budgets
- Measuring fidelity that matters clinically

Q. How to handle high-resolution medical generation that is computationally prohibitive for low-resource clinics?

Challenge 3: Open-Ended Semantics

OUR DIRECTION

Training-free models for open-vocabulary tasks. Long-text clinical descriptions, written the way radiologists actually describe findings, guide the model to segment pathologies zero-shot.

OPPORTUNITIES

- Rare and overlapping pathologies without labelled examples
- New findings added by description, not by retraining

OPEN PROBLEMS YOU COULD OWN

- Grounding long clinical text in noisy ultrasound
- Separating overlapping pathologies in one scan

Q. How do we identify rare or novel pathologies we haven't explicitly trained for?

Challenge 4: Trustworthy System

OUR DIRECTION

We are exploring models that captures uncertainty signals, giving medical segmentation/ detection with uncertainty guarantees.

OPPORTUNITIES

- Confidence on every predicted boundary
- Uncertain cases routed to a radiologist

OPEN PROBLEMS YOU COULD OWN

- Guarantees that still hold under domain shift
- Presenting uncertainty clinicians can act on

None of it matters if clinicians can't trust the output!

Build with Us

- Real Clinical Partners
- Problems that Matter in the Indian Context
- Full Research to Clinic Pipeline
(publications to translational deployment)

Looking for scholars who want to build robust, deployable medical CV from the ground up!

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Thank You!